AgentPulse

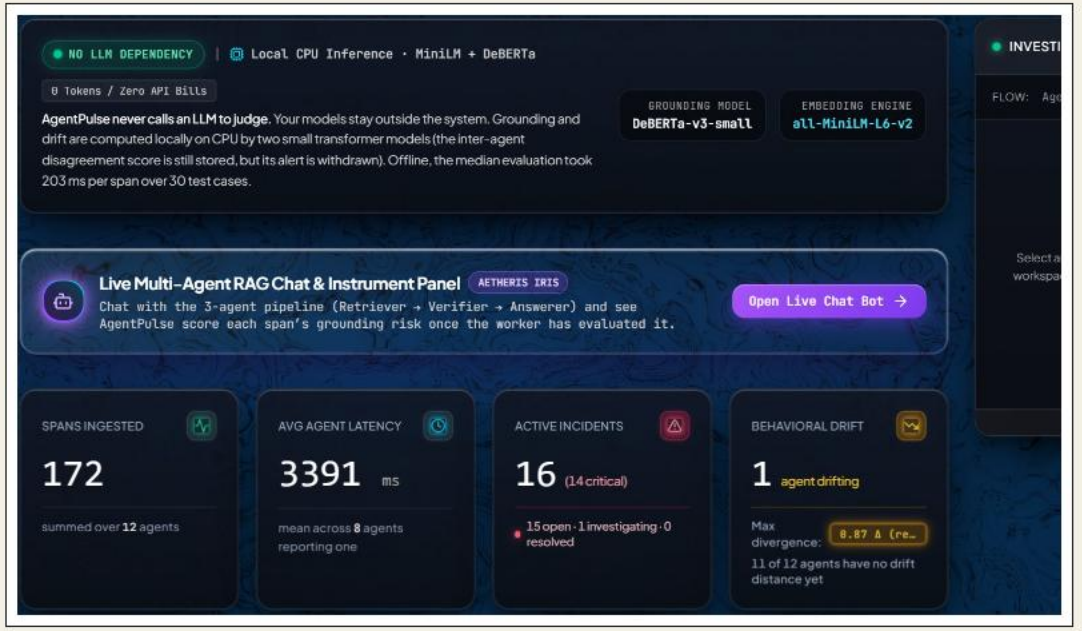
Observability for multi-agent LLM pipelines

LIVE EDITION: AGENTPULSE.CENTRALINDIA.CLOUDAPP.AZURE.COM CODE: GITHUB.COM/SOUM-CODE/AGENTPULSE

Every agent step, checked for grounding and drift, without an LLM judge

Two small local models score each span. Two signals held up on real data, two did not, and this paper says which.

BY SOMNATH REDDY



The deployed console on 1 November 2026. PHOTO: SCREENSHOT OF THE LIVE SITE

THE PROBLEM

In a multi-agent pipeline, a wrong answer looks exactly like a right one



Every call returns 200, nothing throws, latency is normal. The diagram is a schematic.

Grounding failure: an agent says what its evidence does not support.

Drift: an agent slowly answers differently after a change.

Most tools trace cost and latency; quality checks are optional and sampled.

SOURCE *README.md, introduction (tools treat quality evaluation as an optional, sampled add-on).*

THE SYSTEM

AgentPulse scores every span on a CPU, off the agent's critical path



No LLM call anywhere in the evaluator. The SDK adds 0.005 ms per call and never blocks the agent.

SOURCE *README.md, Architecture in short; HOW_IT_WORKS.md (bounded buffer, cooldown 900 s); benchmarks/benchmark_results.json, 2026-08-18 (0.005 ms, P50 over 2,000 calls).*

SIGNAL 1 · GROUNDING

An NLI check with a 0.3 design weight catches contradictions on 68 labelled cases

```
risk = P(contradiction) + 0.3 × P(neutral)
```

EVALUATION SET	CASES	F1	FALSE ALARMS
Test split	30	0.963	0.059
Held-out pool, cut-off 0.7	47	0.927	0.074
Held-out pool, cut-off 0.4	47	0.952	0.074

High risk above 0.7, medium above 0.4. False alarms = share of supported cases flagged (1 of 17; 2 of 27).

Unverifiable: neither supported nor contradicted. A status, not a pass.

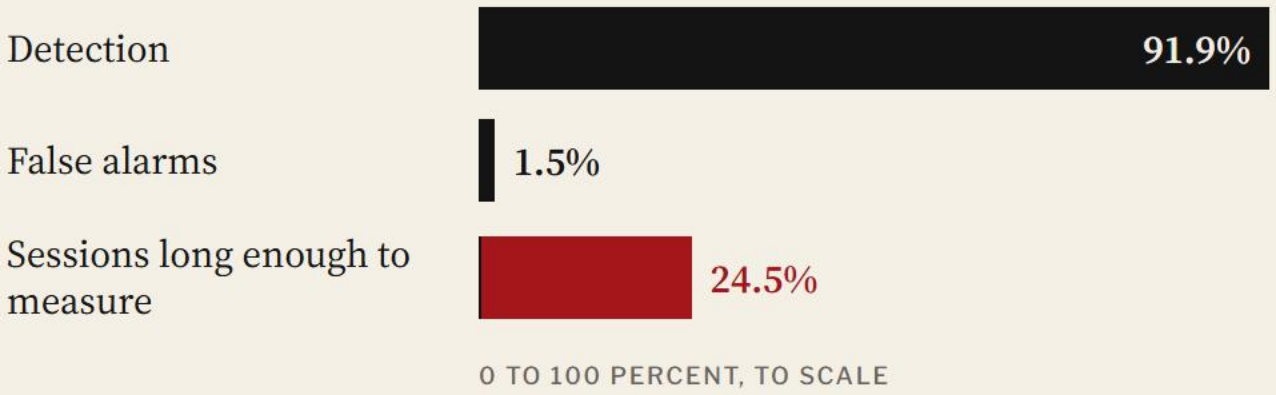
Weights 0 to 0.3 give identical results on our data.

68 cases (39 supported, 29 failures) is a small sample; 5 test cases copy development cases.

SOURCE *GROUNDING_RECALIBRATION_2026-10.md, section 3.6 and tables T3c, T3d (analysis 2026-10-22); HOW_IT_WORKS.md, the verdict rule; backend/app/services/grounding.py (NEUTRAL_RISK_WEIGHT = 0.3).*

RESULT · DRIFT

Drift detection is accurate, but only a quarter of real sessions are long enough to measure



Baseline: an agent's first 20 outputs.
Window: its latest 12.

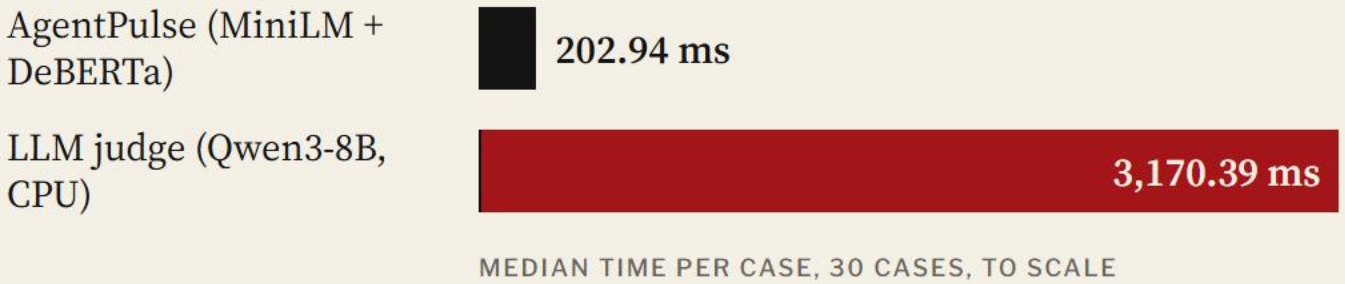
Alert when the two drift apart by a cosine distance above **0.30**.

Offline, real-trace corpus, held-out split, measured once.

SOURCE *DRIFT_REAL_TEXT_DIAGNOSIS_REPORT.md, sections 11.3 and 11.5 (2026-08-27); HOW_IT_WORKS.md (baseline 20, window 12, threshold 0.30).*

RESULT · COST

The evaluator is 15.6 times faster than an LLM judge, at a small accuracy cost



Same 30 cases: judge F1 **1.000**, AgentPulse **0.963**.

One judge model on one machine; not a claim about every judge.

SOURCE *LLM_JUDGE_COMPARISON_REPORT.md, median latency table and correction banner (2026-08-26; F1 at the shipped weight derived from stored probabilities).*

RESULT · VERIFICATION

The live site passed an end-to-end test from outside on 27 October

CHECK ON THE LIVE SITE	RESULT
Health endpoints	200
Key routes, writes without a key	401
Chat turn, three agents on Groq	6.35 s, all scored
Lab hallucination run	HIGH alerts
Every console view, two palettes	0 NaN

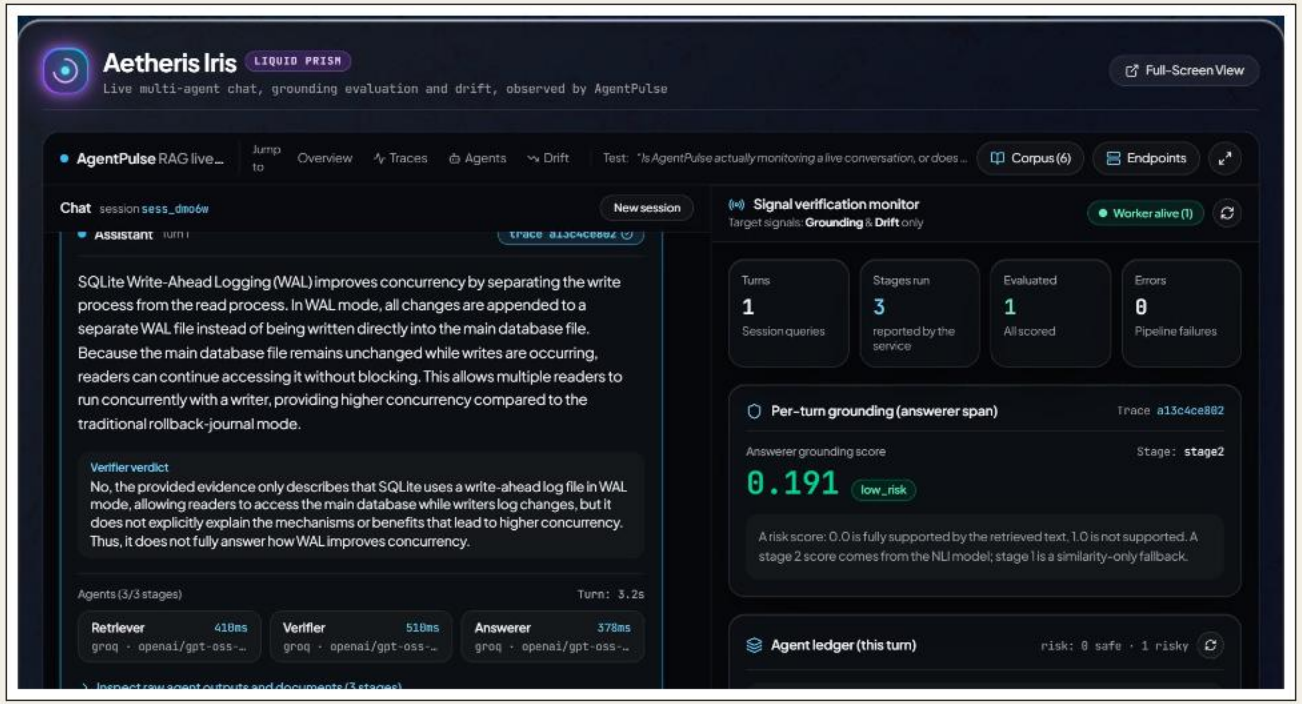
628 backend and **404** dashboard tests pass.

The test also found two real bugs; both were fixed.

SOURCE *docs/E2E_TEST_2026-10-27.md, items 1-25 and findings F1-F3; README.md, Testing and benchmarks (re-run 2026-10-27).*

LIVE DEMO · AGENTPULSE.CENTRALINDIA.CLOUDAPP.AZURE.COM

One question, three agents, and every step scored as it happens



A real turn on 1 November: 3.2 s, all agents on Groq; answerer risk 0.191. A single turn; others were faster or slower. PHOTO: LIVE SITE

1. Ask how SQLite WAL improves concurrency
2. Read the three stages
3. Open the agent ledger
4. Ask an out-of-corpus question
5. Export the trace (key X)

SOURCE *DEMO_GUIDE.md, steps 1-5; screenshot of the live RAG Live view, 1 November 2026.*

ENGINEERING

One small Azure VM, deployed automatically and closed to writes

DEPLOYMENT

One VM, five containers

Azure B2als_v2: 2 vCPU, 4 GB. Docker Compose with TLS.

CI / CD

Tested, then shipped

Every pull request runs the tests. A push to main deploys and rolls back if unhealthy.

SECURITY

Read-open, write-closed

Writes need a password. Key routes return 401. Every public route is rate-limited.

SOURCE *deploy/azure/README.md (VM size); .github/workflows/tests.yml, dashboard.yml, deploy-azure.yml; deploy/azure/Caddyfile; README.md, What works live; SESSION_HANDOFF.md 29.17.2.*

HONEST LIMITS

Two of the four signals failed on real traces, and the evidence base is small

LIMIT	EVIDENCE
Inter-agent disagreement	0 of 10 found
Tool-claim check	F1 0.000
Labelled grounding cases	68
Sessions long enough for drift	24.5%
Deployment	1 VM, 1 SQLite file

Grounding and drift are the two signals that held up.

Both failed alerts were withdrawn, and this slide reports them.

SOURCE *README.md, Known limits; COMPETITIVE_POSITIONING.md 5.2; TOOL_CLAIM_EXTERNAL_TEST_REPORT.md section 10; GROUNDING_RECALIBRATION_2026-10.md; DRIFT_REAL_TEXT_DIAGNOSIS_REPORT.md 11.5.*

CONCLUSIONS

Conclusions

1. Every span can be scored on a CPU without an LLM judge: **202.94 ms** per case.
2. Grounding (F1 **0.963**) and drift held up; two other signals failed and are withdrawn.
3. It is live, tested (**628 + 404**) and open to examiners.

Next: more labelled data, a server database, test-gated deploys.

SOMNATH REDDY

`agentpulse.centralindia.cloudapp.azure.com`
`github.com/Soum-Code/agentpulse`

SOURCE *LLM_JUDGE_COMPARISON_REPORT.md; GROUNDING_RECALIBRATION_2026-10.md T3c; README.md, Testing and benchmarks (2026-10-27).*